

Enhancing Covid-19 detection with YOLOv6 object detection algorithm in CT Scan images

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Abstract

The COVID-19 pandemic has severely disturbed global healthcare systems, laying more stress on the need for more efficient diagnostic methods. In response to this, computer-aided diagnosis (CAD) has become instrumental in improving diagnostic speed and accuracy, providing tremendous support to medical professionals in identifying COVID-19 cases. My task focuses on detecting and classifying abnormalities in CT scan images. We utilize a dataset provided by medical dataset platforms and applied the YOLOv6 model for object detection. In the previous studies section, I walk you through various object detection techniques, which are broadly classified into one-stage and two-stage models. Notable examples of these include the Faster R-CNN and the YOLO series. In the methodology section, I present a detailed overview of the YOLOv6 model. In section IV, I compare experimental results, using mean average precision (mAP) as the evaluation metric. A higher mAP value indicates better model performance. The mAP@0.5 achieved by our YOLOv6 model is 0.731, which is 0.108 higher than YOLOv5, respectively, demonstrating its superior performance.

Keywords: COVID-19, Computer-aided diagnosis, CT scan, detection, YOLO, R-CNN

1. Introduction

The global outbreak of COVID-19 has resulted in unmatched challenges to healthcare systems worldwide, highlighting the need for efficient diagnostic tools to support timely treatment. [1] General COVID-19 diagnosis includes PCR (polymerase chain reaction) and antigen tests which cost several hours.[3] If we are in need of immediate results we can make use of radiographs of the chest which can be obtained in a short amount of time. [2] Since COVID-19 and other bacterial infections in the chest like pneumonia, bronchitis looks more identical in radiographs, it makes it more difficult for the radiologists for diagnosis. With the help of Computer aided Diagnosis the efficiency of the diagnosis increases exponentially, which helps doctors to come to conclusion quickly and start the diagnosis. This also plays an important role in COVID-19 detection. In this paper I focus on classifying CT-Scan images which mainly contains two kinds i.e. COVID and NON-COVID. In further I have done study on the previous and related works, used models for this experiment, and result comparison. Finally I concluded based on these factors and proposed further enhancement for the model.

2. Previous Studies

In recent years, [4] numerous approaches to object detection have been presented, which can be divided into two general categories. Modern deep learning-based object detection algorithms are often separated into two-stage and single-stage techniques, depending on the process of generating candidate boxes [5].

2.1. Two-stage method

Traditional object detection methods frequently suffered from low precision. To solve this, Girshick et al. [6] proposed R-CNN, a pioneering deep learning-based object detection system. This paved the way for further improvements. Ren et al. [7] presented Faster R-CNN, which improved on the prior method by using a Region Proposal Network (RPN) to produce candidate boxes. The RPN provided scale-specific anchors to replace traditional methods like selective search, enabling end-to-end network training while enhancing computational efficiency. The Faster R-CNN architecture consists of convolutional layers, an RPN network, regions of interest (ROI), and classification and regression layers.

2.2 One-stage method

While two-stage approaches are extremely precise, they are often inefficient for real-time applications. To address this issue, Redmon et al. [8] developed YOLO (You Only Look Once) v1, which eliminates the candidate box extraction stage. Instead, it combined feature extraction, classification, and regression into a single convolutional network. This simplified the network structure while greatly increasing detection speed from 7 FPS in Fast R-CNN to 45 FPS in YOLOv1. The discovery of YOLO prompted the development of single-stage object detection algorithms, which are now often utilized in real-time applications.

2.3 Self contribution

- I have used the YOLOv6 model for object detection, which improves on the YOLO series by increasing detection speed and accuracy.
- I present my dataset, assess its properties, and use augmentation strategies to improve model performance.
- In my experiments, I compare YOLOv6 to its previous model and show that it performs better in terms of speed and mean Average Precision (mAP).

3. Methodology

The entire structure of YOLOv6 is intended to improve the accuracy and speed of object detection. It uses advanced backbone and neck structures to accurately detect COVID-19 infection regions in CT scan pictures. Below is the explanation of the architecture:

3.1. Backbone Structure

YOLOv6's core network processes input CT scan images and extracts critical features, allowing for rapid and precise COVID-19 detection. To boost performance, YOLOv6 employs EfficientRep Backbone, an updated version of the CSPNet architecture used in YOLOv5.

EfficientRep Backbone: The basic component of YOLOv6's backbone employs a lightweight and extremely efficient feature extraction method that blends depthwise convolutions and CSPNet. This approach optimizes both feature extraction and gradient flow, decreasing

redundancy and minimizing FLOPs (floating-point operations per second), which are critical for processing large-scale CT images in real time. The backbone is designed to extract detailed information from CT scan images, allowing for better identification of infection locations.

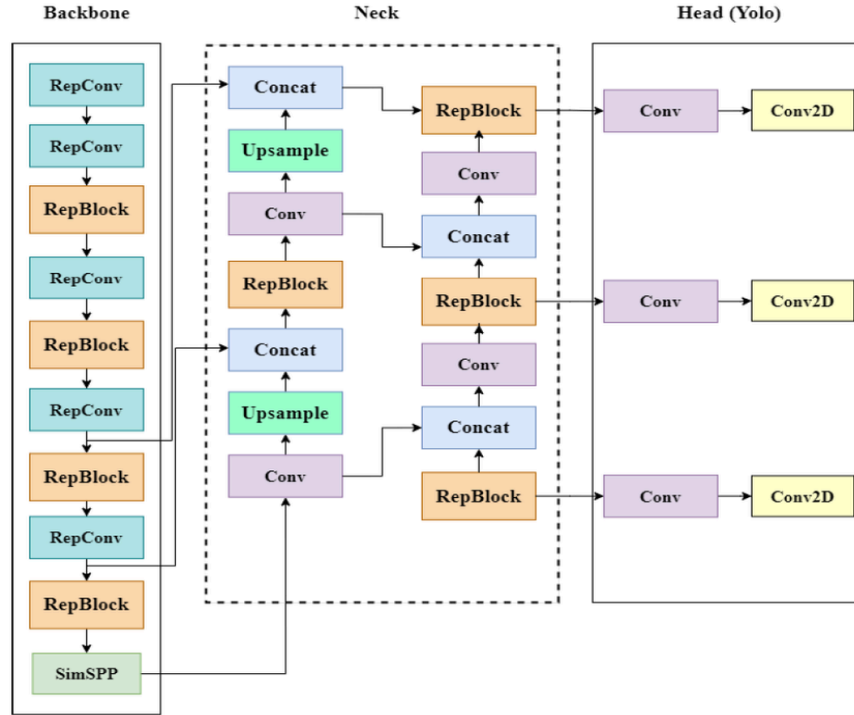


Figure1: Architecture of YOLOv6

3.2. Neck Structure

The neck in YOLOv6 is in charge of collecting multi-scale information collected from the backbone, hence improving the model's capacity to recognize objects of various sizes. This is especially important in medical imaging, since COVID-19 infection spots on CT images might present at varying scales.

YOLOv6 uses Rep-PANet (Path Aggregation Network), an optimized variant of PANet, to increase feature fusion. It improves both bottom-up and top-down information flow, allowing for more precise detection of small infection zones. The network transmits multi-scale features from lower to higher layers and vice versa, guaranteeing that detailed local features, such as small COVID-19 infection zones, are combined with global context information. It is also responsible for Multi-scaling feature fusion and Adaptive feature polling.

3.2. Head Structure

The head of YOLOv6 handles the final prediction task, including bounding box regression and object classification. YOLOv6 uses a decoupled detection head to separate the processes of classification and localization, improving the overall detection accuracy.

Key components of YOLOv6:

1. **RepVGG Block** : This block streamlines the network architecture during inference by combining batch normalization (BN) and convolution layers.
2. **EfficientRep Block** : YOLOv6 makes use of EfficientRep blocks, which combine depthwise convolution and CSPNet's partial feature reuse approach to optimize feature extraction while reducing computing costs.
3. **Spatial Pyramid Pooling** : YOLOv6 uses Spatial Pyramid Pooling (SPP) layers for multi-scale feature fusion, performing max-pooling at several scales (e.g., 1×1 , 3×3 , 5×5 , 7×7), as seen in YOLOv5.

4. Experimental Results

The training dataset comprises 1256 chest radiograph data in DICOM format. The scans were then converted to png format and labeled and annotated using roboflow. In detecting objects YOLO used IOU (Intersection Over Union).

$$IOU = \frac{Area\ of\ Overlap}{Area\ of\ Union}$$

where Area of overlap is the overlapping of two bounding boxes and the area of union indicates the overall area covered by both ground truth and predicted box frames.

4.1. Result analysis

Below are some input images that I have used to train the model and the output that I have obtained from the model and their properties are:

Height	Width	Batch_size	EPOCHS
512	512	16	50

Table 1: Properties of image

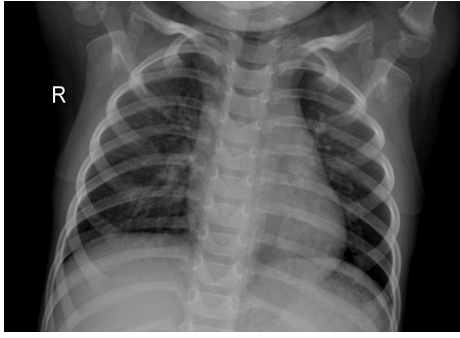


Figure2: Input Figures

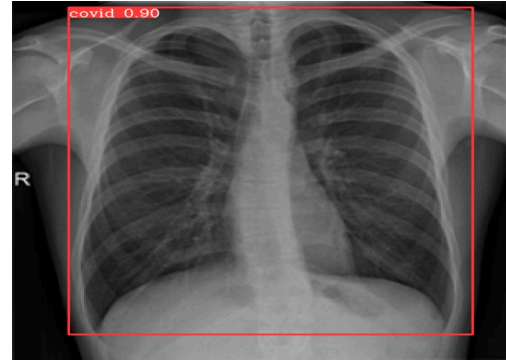
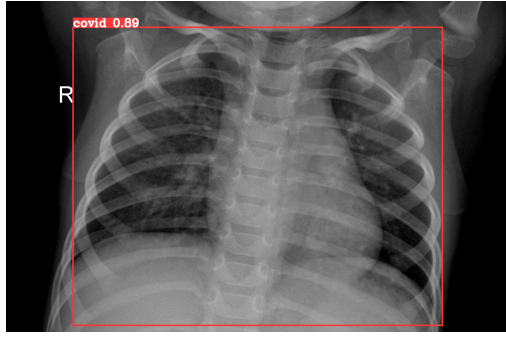


Figure3: output Figures

I focused on finding Mean Average Precision (mAP) score [9] for the trained model. To compare the performance of the model I choose its previous version i.e. YOLOv5 and used the same metric mAP and same dataset to gather its accuracy. The following is the accuracy of both models that I have obtained.

Models	mAP score
YOLOv6	0.731
YOLOv5	0.623

Table 2: mAP scores comparison

As we can see YOLOv6 is 0.108 higher than YOLOv5 respectively.

5. Conclusion

This section concludes our work on abnormality detection in lung ct scan. Our task focuses on detecting and classifying abnormalities [10]. We used the open source medical data and YOLOv6 model. In the previous studies section, we discuss approaches for objective detection, including one-stage and two-stage objection detection. In Section III, we discuss the YOLOv6 model in full.

Comparing the experiment and findings are presented in section IV. Our trials use mean average precision (mAP) as a statistic, with larger values indicating better results for the model. YOLOv6 has a mAP of 0.731, which is 0.108 greater than YOLOv5 respectively.

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